

1. Administrative & Study Identification

Full Study Title: A Randomised, Double-blind, Placebo-controlled, Multi-centre Phase 2b Study to Evaluate the Efficacy, Safety and Tolerability of AZD2693 in Participants With Non-cirrhotic Non-alcoholic Steatohepatitis (NASH) With Fibrosis Who Are Carriers of the PNPLA3 rs738409 148M Risk Allele **Short Title/Acronym:** FORTUNA **Protocol Number:** D7830C00004 **NCT Number:** NCT05809934 **Secondary IDs:** 2022-001629-65 (EudraCT Number), 2023-509704-14-00 (Registry Identifier) **Sponsor Name:** AstraZeneca **Investigational Product:** AZD2693 (PNPLA3 antisense oligonucleotide) **Phase of Development:** Phase 2b (Phase II) **Medical Monitor:** AstraZeneca Clinical Operations Lead

2. Study Synopsis & Rationale

2.1 Study Objective Primary Objective: To assess the efficacy, safety, and tolerability of AZD2693, primarily evaluating the histological resolution of Metabolic Dysfunction-Associated Steatohepatitis (MASH/NASH) without the worsening of fibrosis. **Secondary Objectives:** To evaluate the improvement in liver fibrosis, reduction in relative liver fat content, changes in PNPLA3 mRNA expression, and changes in inflammatory biomarkers.

2.2 Background & Rationale The PNPLA3 rs738409 148M risk allele is the strongest known genetic determinant linked to the progression of liver steatosis, inflammation, and fibrosis leading to cirrhosis. AZD2693 is a potent GalNAc-conjugated antisense oligonucleotide (ASO) designed to target and silence PNPLA3 mRNA in the liver. This study aims to evaluate its potential as a targeted, precision-medicine therapeutic approach to resolve MASH and improve fibrosis in genetically predisposed patients.

3. Study Design & Methodology

3.1 Design Framework Study Type: Interventional **Control Type:** Placebo-controlled **Blinding:** Double-blind **Randomization:** Randomised (Multi-arm: AZD2693 Dose 1, AZD2693 Dose 2, Placebo)

3.2 Planned Enrollment & Duration Target \$N\$: 180 participants **Number of Sites:** 165 locations (Global/Multi-centre) **Estimated Study Start:** 14-Mar-2023 **Estimated Primary Completion:** 14-Jul-2025 **Estimated Study Completion:** 28-Sep-2025

4. Subject Selection (Eligibility Criteria)

4.1 Inclusion Criteria 1. Participant must be 18 to 75 years of age (inclusive) at the time of signing the informed consent. **Sex:** All. **Healthy Volunteers:** No. 2. Participants who are carriers for the PNPLA3 rs738409 148M risk allele. 3. Participants with histological evidence of NASH based on central pathologist evaluation of a liver biopsy obtained up to 6 months before randomisation: Definitive NASH with NAS ≥ 4 (with ≥ 1 in each component: steatosis, lobular inflammation, and ballooning) and F2 or F3 fibrosis.

4.2 Exclusion Criteria 1. Evidence of cirrhosis or other significant liver diseases (e.g., viral hepatitis, autoimmune hepatitis). 2. History of hepatic decompensation or hepatocellular carcinoma, including ascites, hepatic encephalopathy, or variceal bleeding. 3. Concomitant use of prohibited medications (e.g., strong CYP inhibitors) or active involvement in another interventional clinical trial within the wash-out period. 4. Historical persistent or pre-existing renal disease marked by $\text{eGFR} < 40 \text{ mL/min/1.73 m}^2$. 5. Confirmed platelet count outside the normal range at the screening visit.

5. Intervention & Treatment Plan

5.1 Investigational Regimen Dosage/Frequency: Two different doses of AZD2693 (Dose 1 and Dose 2) vs. matching Placebo. **Route of Administration:** Subcutaneous (SC) injection. **Duration of Treatment:** 52 weeks.

5.2 Concomitant & Prohibited Therapy Permitted: Standard of care therapies for concurrent co-morbidities such as type 2 diabetes, obesity, and hypertension, provided patients remain on stable doses. **Prohibited:** Other investigational drugs and specifically prohibited metabolism-altering or liver-targeted agents outside the study parameters.

6. Study Timeline & Visit Schedule

Visit ID	Window (Days)	Key Activities/Procedures	:---
:---	:---	Screening	Day -28 to 0 Informed Consent, Genotyping, Baseline Liver Biopsy, MRI-PDFF
Baseline	Day 0	Randomization, First SC Dose, Vital Signs	Interim Weeks 12, 24, 36 Safety Labs, Efficacy Assessment (MRI-PDFF, Biomarkers), PK/PD sampling
End of Study	Week 52	Final Onsite Visit, Final Liver Biopsy, Exit Interview, IP Accountability	

7. Outcome Measures (Endpoints)

7.1 Primary Endpoint Definition: Histological resolution of NASH/MASH without worsening of fibrosis at 52 weeks. **Measurement Method:** Central pathologist evaluation of liver biopsy.

7.2 Secondary & Exploratory Endpoints 1. Improvement in liver fibrosis by at least one stage without worsening of NASH/MASH. 2. Reduction in relative hepatic fat content, evaluated via MRI-PDFF. 3. Changes in circulating levels of target engagement biomarkers (e.g., PNPLA3 mRNA) and inflammatory markers (e.g., hsCRP).

8. Safety & Adverse Event Reporting

Adverse Event (AE) Monitoring: Standard collection via patient reporting and clinical assessment (with specific focus on injection-site reactions and hepatotoxicity). **Serious Adverse Events (SAEs):** Reported within 24 hours to the Sponsor. **Data Monitoring Committee (DMC):** Independent external Data Monitoring Committee to oversee safety data throughout the trial phase.

9. Statistical Analysis Plan (SAP) Summary

Analysis Populations: Intent-to-Treat (ITT) population for primary efficacy endpoints; Safety Set for AE/SAE evaluations. **Sample Size Justification:** $N=180$ powered ($\alpha = 0.05$, $\text{Power} \geq 80\%$) to detect a statistically significant difference in MASH resolution rates between AZD2693 groups and placebo. **Handling of Missing Data:** Non-responder imputation (NRI) for binary histological endpoints; Multiple Imputation (MI) and Mixed-Effects Model for Repeated Measures (MMRM) for continuous data.

10. Quality Assurance & Regulatory Compliance

GCP Compliance: This study will be conducted in accordance with Good Clinical Practice (GCP) and the Declaration of Helsinki. **Monitoring Plan:** Scheduled onsite and remote monitoring visits by Clinical Research Associates (CRAs) to ensure protocol adherence. **Audit Trail:** Secure, validated Electronic Data Capture (EDC) system with full audit trails, eCRF locking, and formalized data clarification processes.

11. Study Identifiers & Governance

Primary ID: D7830C00004 **Registry Number:** NCT05809934 **Secondary IDs:** 2022-001629-65 (EudraCT Number), 2023-509704-14-00 (Registry

Identifier) **Sponsor/Lead Organization:** AstraZeneca **Study Title:** FORTUNA **Official Regulatory Title:** A Randomised, Double-blind, Placebo-controlled, Multi-centre Phase 2b Study to Evaluate the Efficacy, Safety and Tolerability of AZD2693 in Participants With Non-cirrhotic Non-alcoholic Steatohepatitis (NASH) With Fibrosis Who Are Carriers of the PNPLA3 rs738409 148M Risk Allele

12. Clinical Framework (NASH/MASH Specific)

Primary Condition: Nonalcoholic Steatohepatitis (NASH) / Metabolic Dysfunction-Associated Steatohepatitis (MASH) **Diagnosis Threshold:** Definitive NASH (NAS $\geq$ 4), F2-F3 Fibrosis, PNPLA3 148M risk allele carrier **Medical Methodology:** Targeted antisense oligonucleotide therapy (GalNAc-ASO) **Optimization Target:** Hepatic PNPLA3 mRNA knockdown and MASH resolution

13. Study Procedures & Methodology

Management Pathway: Clinical pathway for targeted NASH/MASH therapy evaluation **Education Protocol (HCP):** Protocol-specific training for subcutaneous injections, liver biopsy tissue handling, and AE monitoring **Education Protocol (Patient):** Onsite training for subcutaneous injection procedures and general lifestyle/dietary counseling **Quality Audit Mechanism:** Centralized pathology reading for all biopsies and independent central reviews for imaging (MRI-PDFF)

14. Observation & Follow-Up Schedule

Total Duration: 52 weeks (plus subsequent safety follow-up) **Onsite Visit Cadence:** Baseline, Weeks 4, 12, 24, 36, 52 **Remote Monitoring Frequency:** Every 4 weeks (Q4W) **Checklist Review Interval:** Every 12 weeks (Q12W)

15. Sign-off & Approval

Role	Name	Signature	Date	:---	:---	:---	:---
Principal Investigator	[Name]	_____	[DD-MMM-YYYY]				
Clinical Operations Lead	[Name]	_____	[DD-MMM-YYYY]			Lead	
Statistician	[Name]	_____	[DD-MMM-YYYY]				